REGRESSION AND CLASSIFICATION IN MACHINE LEARNING

Understanding Types, Equations, Graphs, and Applications

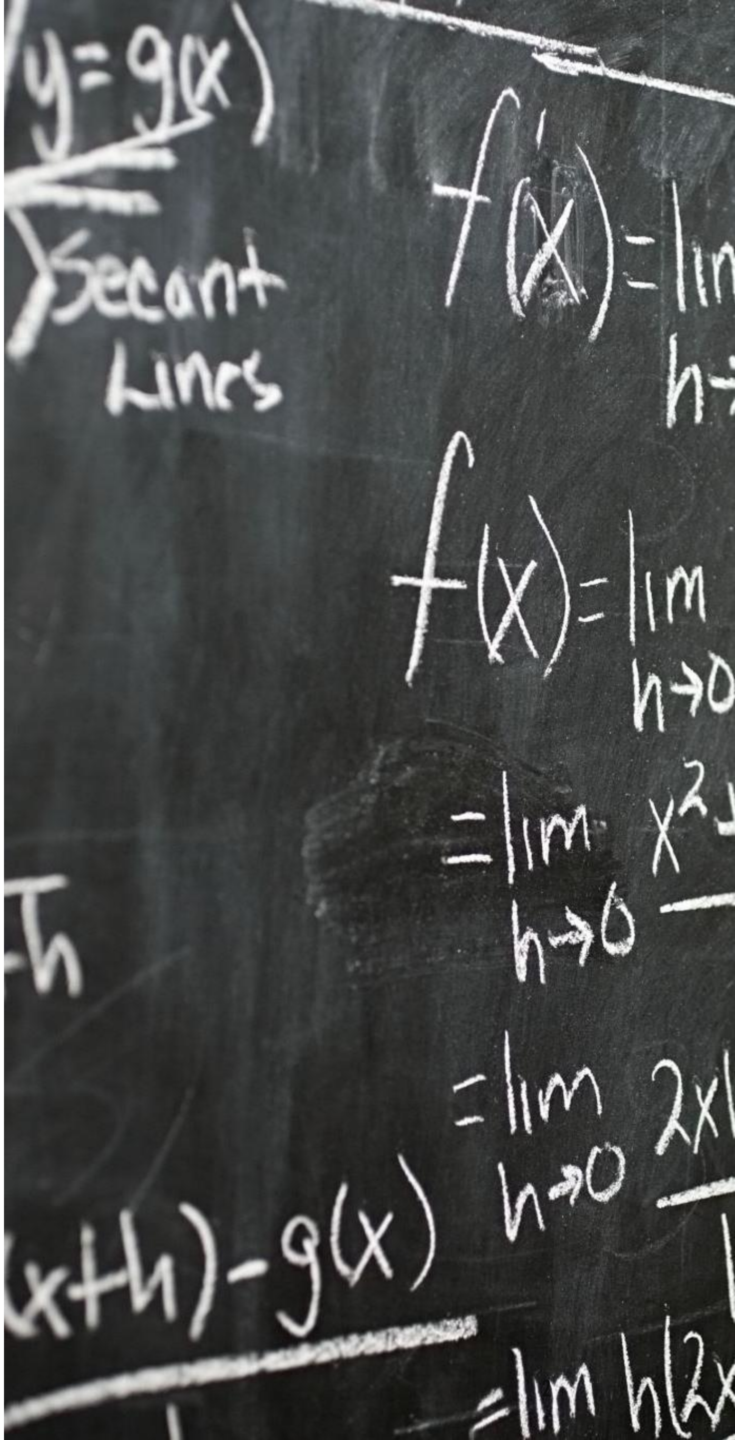
Your Name

Date

INTRODUCTION TO REGRESSION

Definition: Regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables.

Purpose: To predict the value of the dependent variable based on the values of the independent variables.



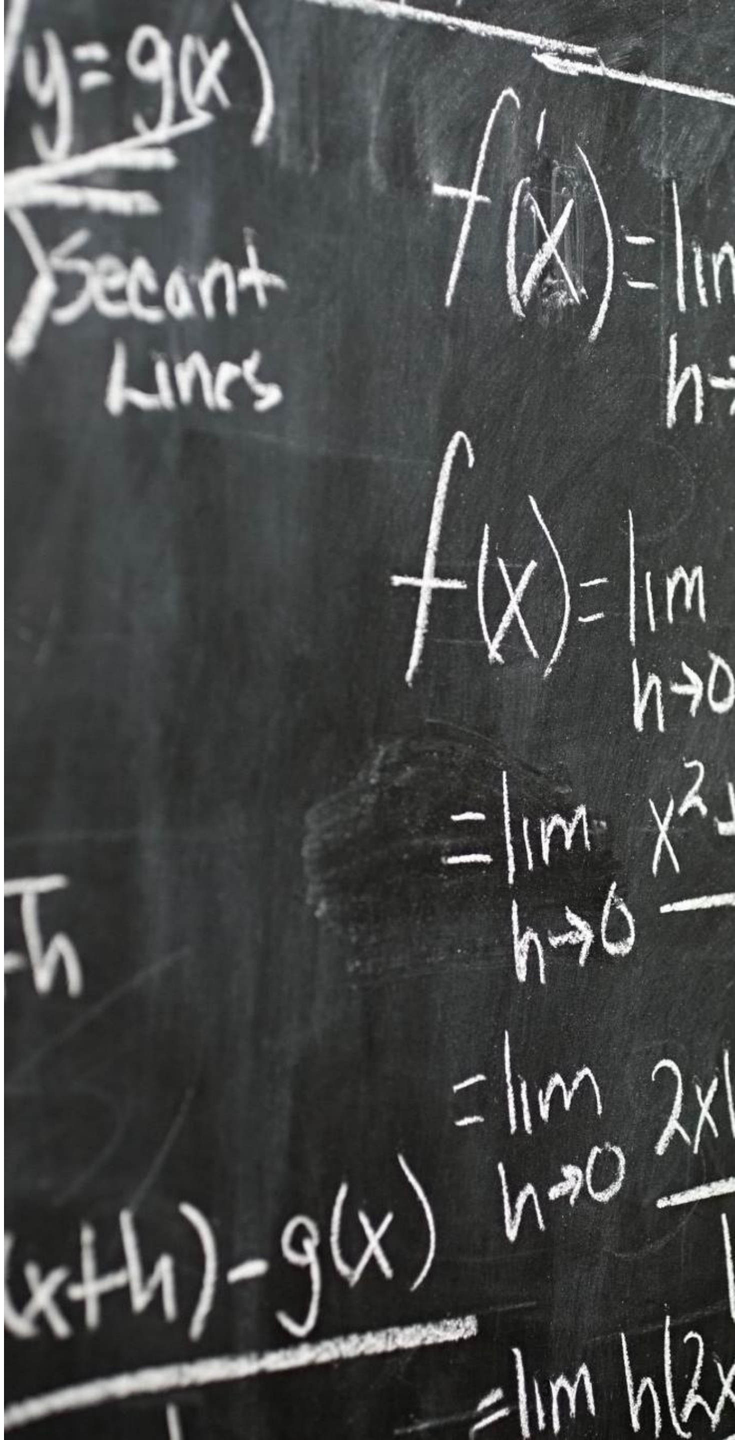
TYPES OF REGRESSION

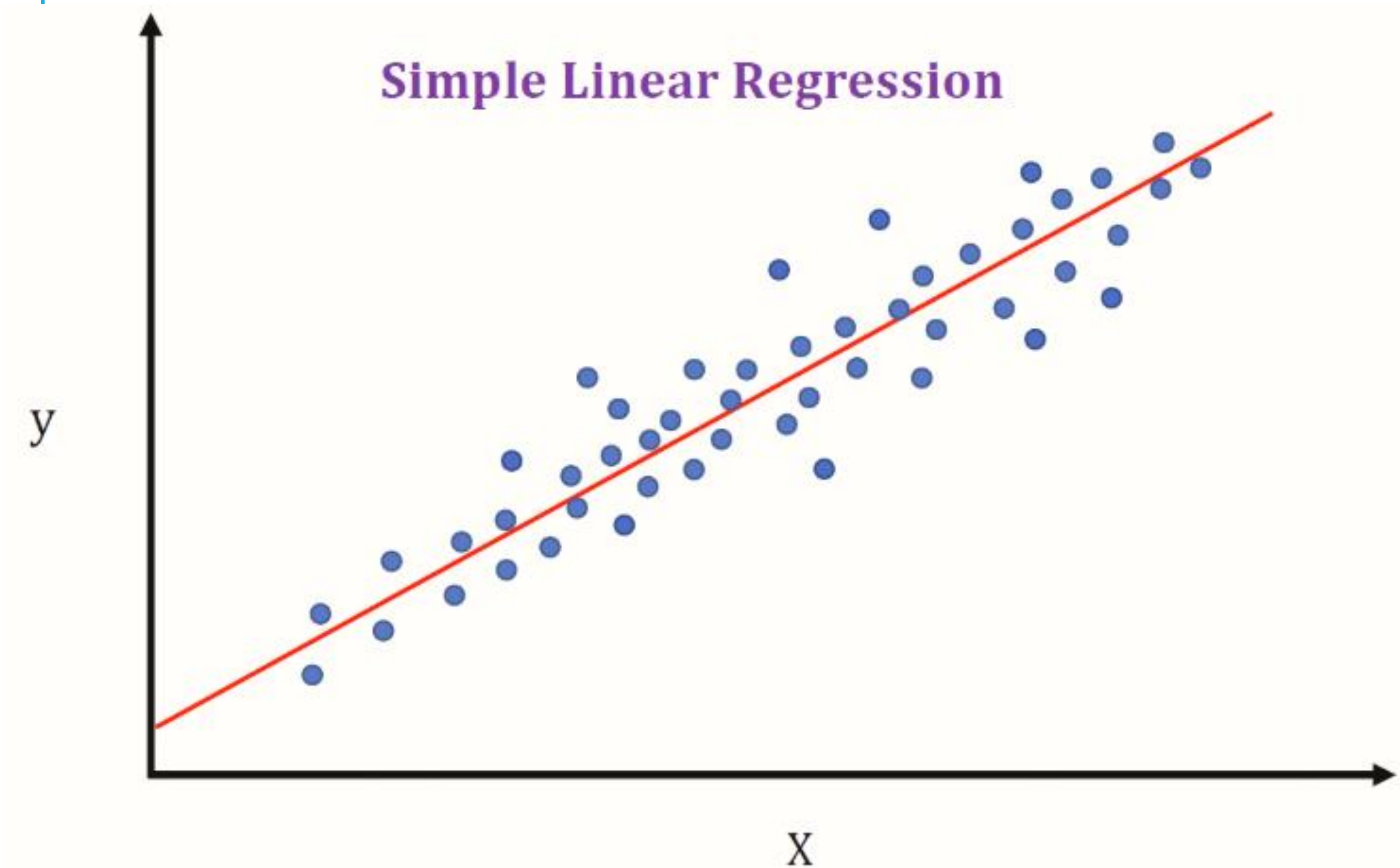
Linear Regression: Models the relationship between two variables by fitting a linear equation to observed data.

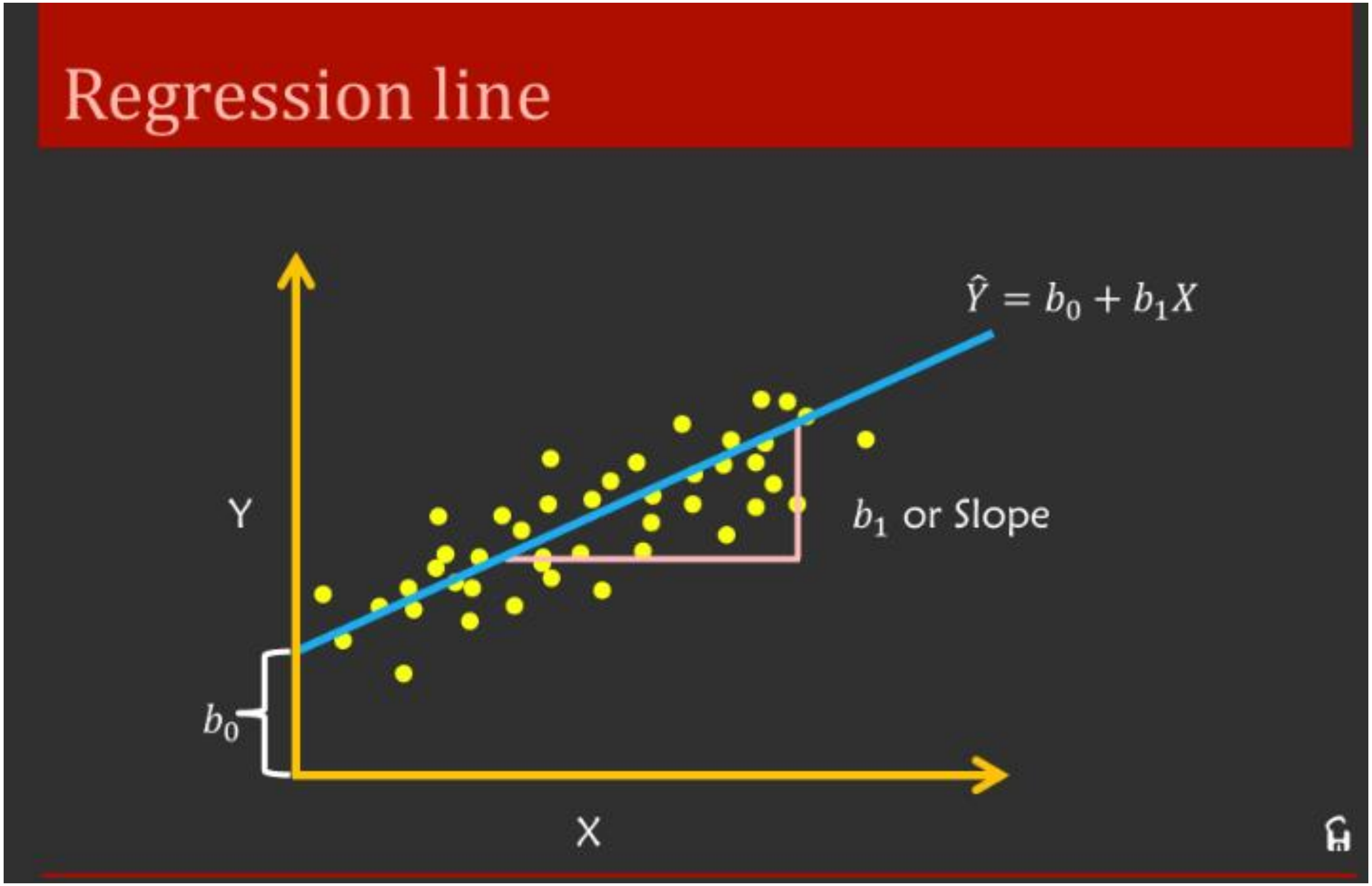
Multiple Linear Regression: Extends linear regression by using multiple independent variables.

Polynomial Regression: Models the relationship between variables as an nth degree polynomial.

Logistic Regression: Used for binary classification problems.

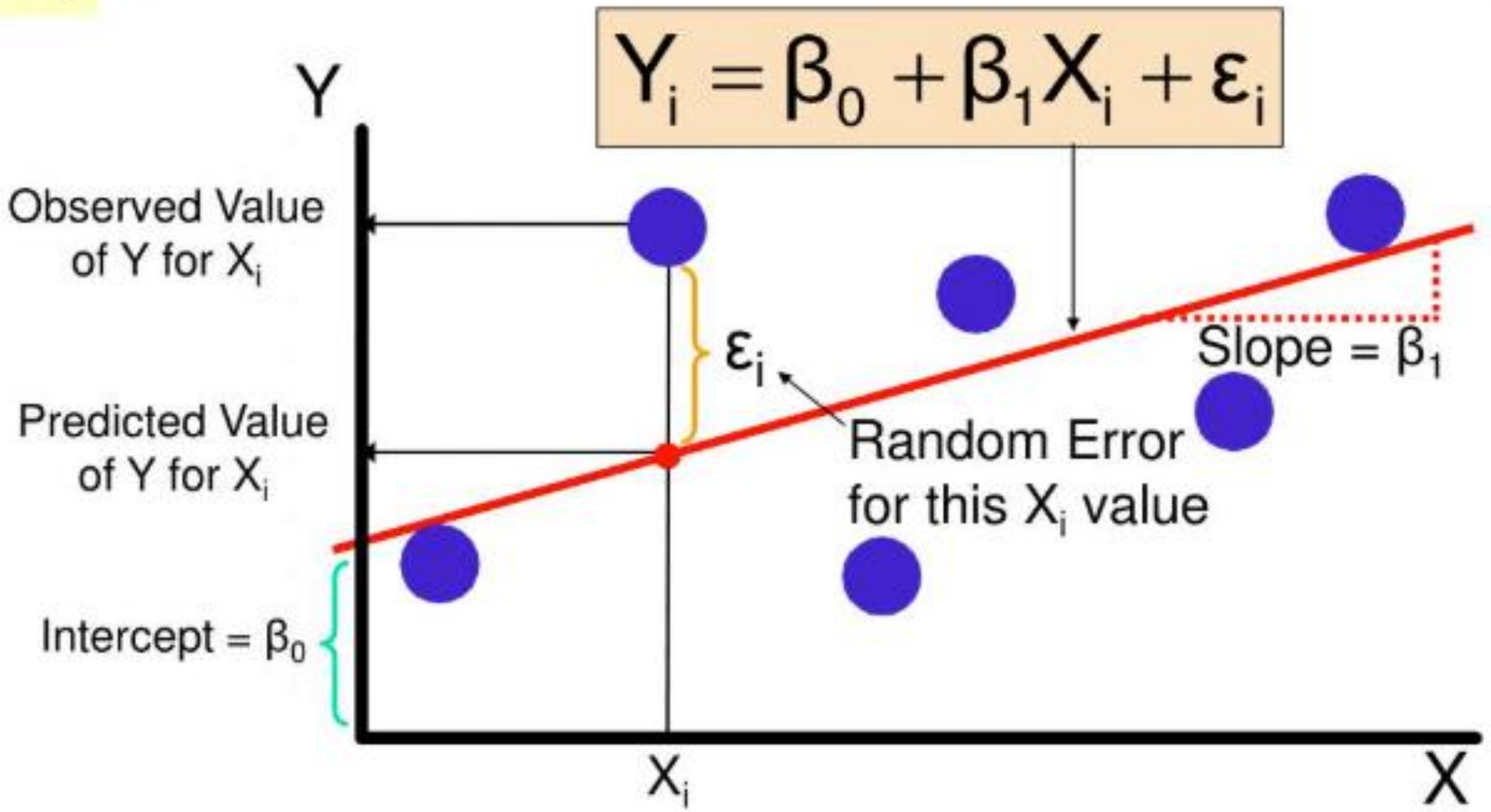



Simple Linear Regression
y
X


Regression line
$\hat{Y} = b_0 + b_1 X$
Y
b_1 or Slope
b_0
X

Simple Linear Regression Model

(continued)

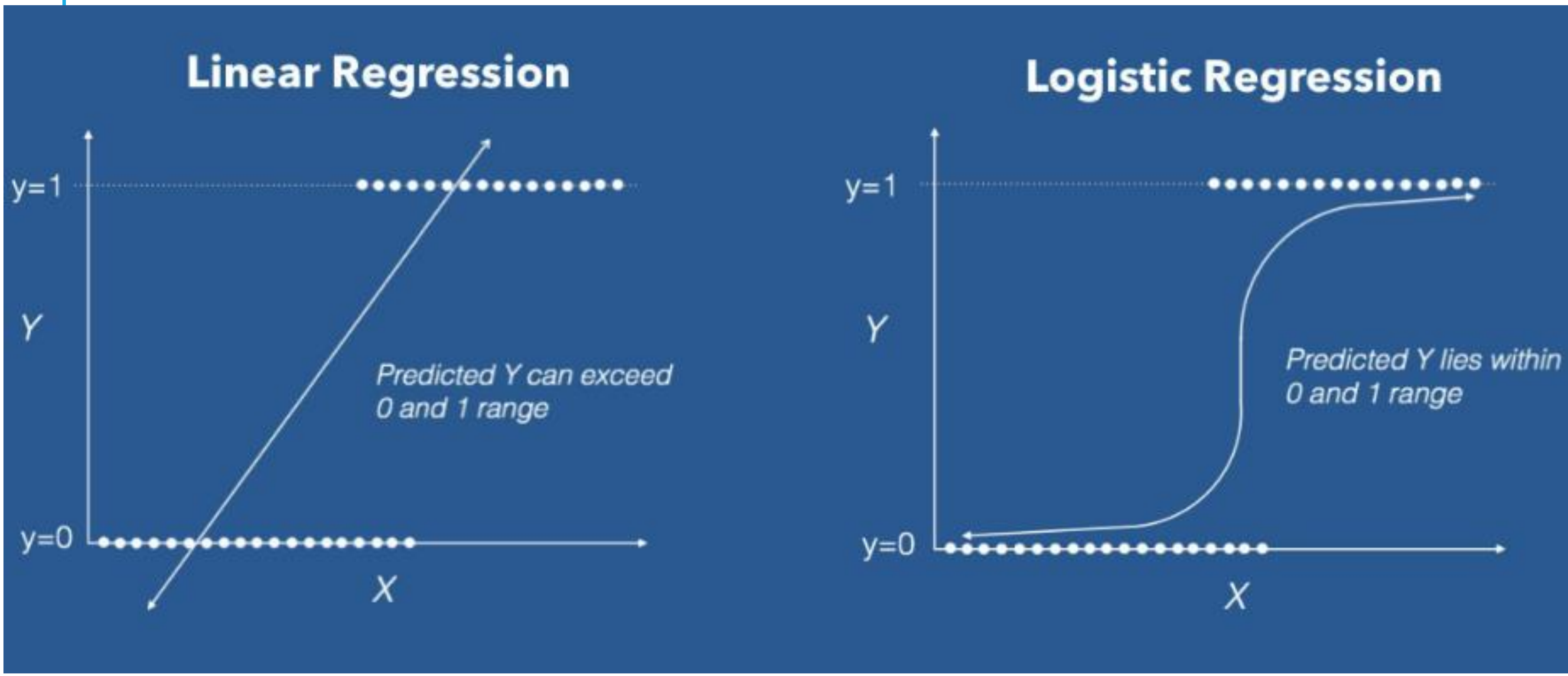


REGRESSION EQUATIONS

Linear Regression Equation: $y = \beta 0 + \beta 1x + \varepsilon$

Multiple Linear Regression Equation: $y = \beta 0 + \beta 1x1 + \beta 2x2 + \ldots + \beta nxn + \varepsilon$

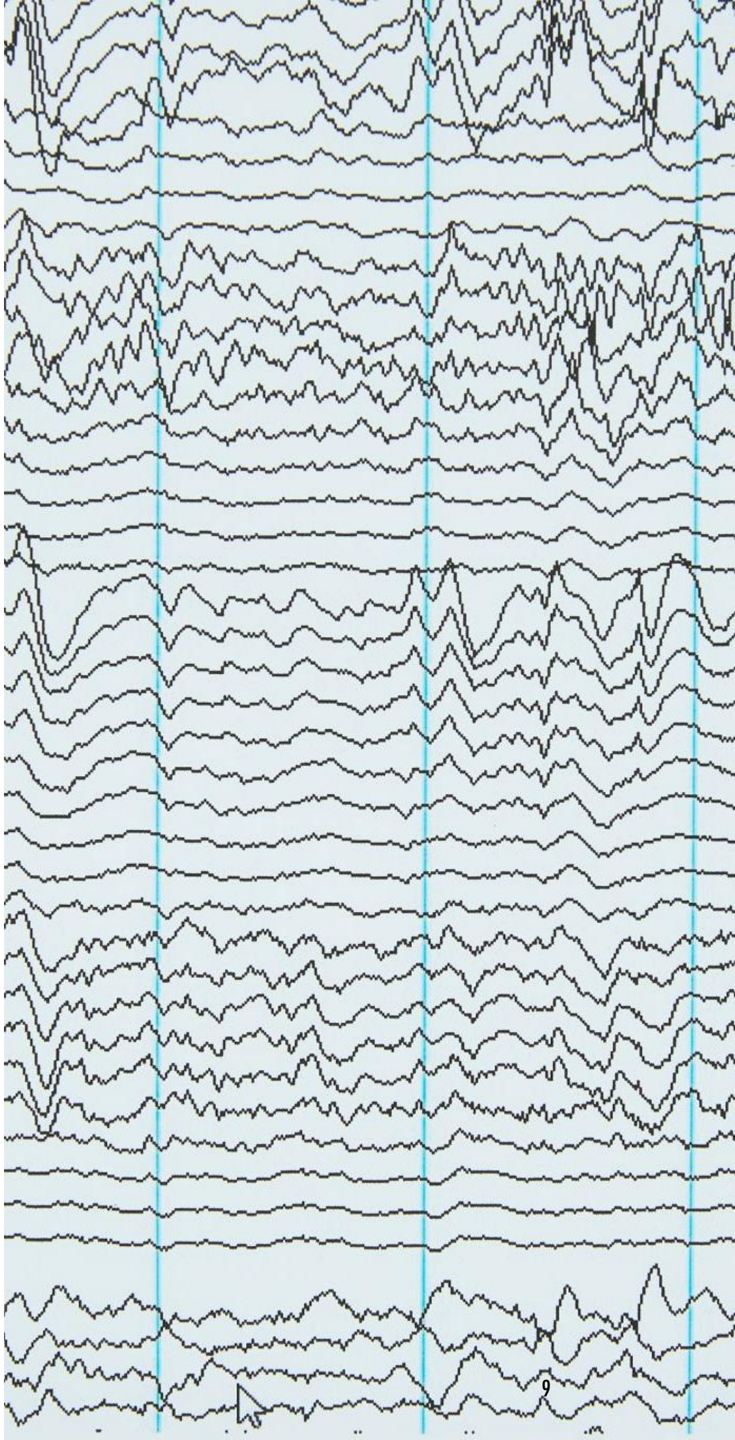
Logistic Regression Equation: $\text{logit}(p) = \log(p/(1-p)) = \beta 0 + \beta 1x$


Linear Regression
y=1
Y
y=0
X
Predicted Y can exceed
0 and 1 range
Logistic Regression
y=1
Y
y=0
X
Predicted Y lies within
0 and 1 range

GRAPHS OF REGRESSION MODELS

Linear Regression Graph: Show a scatter plot with a fitted linear line.

Logistic Regression Graph: Show a sigmoid curve for binary classification.



EXAMPLES OF REGRESSION

- Linear Regression Example: Predicting house prices based on square footage.
- Multiple Linear Regression Example: Predicting sales based on advertising spend across different media.
- Polynomial Regression Example: Modeling the growth rate of a plant over time.
- Logistic Regression Example: Predicting whether an email is spam or not.

RELEVANCE IN MACHINE LEARNING

Predictive Modeling: Regression is widely used for predictive modeling in various domains.

Classification Problems: Logistic regression is a fundamental method for binary classification.

VIDEO LINKS

https://www.youtube.com/watch?v=FLJ0yYetywE&t=549s

https://www.youtube.com/watch?v=CtsRRUddV2s&t=99s

https://www.youtube.com/watch?v=feDJkDaNuOk&t=748s

STEPS TO USE LINEAR REGRESSION IN ML

Step 1: Import Libraries

First, you need to import the necessary libraries.

```
import numpy as np

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

import matplotlib.pyplot as plt
```

Step 2: Load the Dataset

Load your dataset into a pandas DataFrame.

```
# Example: Load a CSV file
data = pd.read_csv('your_dataset.csv')
```

Step 3: Explore the Data

Explore your data to understand its structure and contents.

```
print(data.head())
print(data.describe())
print(data.info())
```

Step 4: Preprocess the Data

Handle missing values, encode categorical variables, and split the data into features (X) and target (y).

```
# Example: Drop rows with missing values
data = data.dropna()

# Example: Select features and target
X = data[['feature1', 'feature2', 'feature3']]  # Replace with your feature columns
y = data['target']  # Replace with your target column
```

Step 5: Split the Data

Split the data into training and testing sets.

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
```

Step 6: Train the Model

Create a Linear Regression model and train it on the training data.

```
model = LinearRegression()

model.fit(X_train, y_train)
```

Step 7: Make Predictions

Use the trained model to make predictions on the test data.

```
y_pred = model.predict(X_test)
```

Step 8: Evaluate the Model

Evaluate the model's performance using metrics like Mean Squared Error (MSE) and R-squared (R^2).

```
mse = mean_squared_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print(f'Mean Squared Error: {mse}')

print(f'R-squared: {r2}')
```

Step 9: Visualize the Results

Visualize the actual vs. predicted values.

```
plt.scatter(y_test, y_pred)
plt.xlabel('Actual Values')
plt.ylabel('Predicted Values')
plt.title('Actual vs. Predicted Values')
plt.show()
```

STEPS TO USE LOGISTIC REGRESSION IN ML USING PYTHON

Step 1: Import Libraries

First, you need to import the necessary libraries.

```
import numpy as np

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report

import matplotlib.pyplot as plt

import seaborn as sns
```

Step 2: Load the Dataset

Load your dataset into a pandas DataFrame.

```
# Example: Load a CSV file
data = pd.read_csv('your_dataset.csv')
```

Step 3: Explore the Data

Explore your data to understand its structure and contents.

```
print(data.head())

print(data.describe())

print(data.info())
```

Step 4: Preprocess the Data

Handle missing values, encode categorical variables, and split the data into features (X) and target (y).

```
# Example: Drop rows with missing values
data = data.dropna()

# Example: Select features and target
X = data[['feature1', 'feature2', 'feature3']]  # Replace with your feature columns
y = data['target']  # Replace with your target column
```

Step 5: Split the Data

Split the data into training and testing sets.

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Step 6: Train the Model

Create a Logistic Regression model and train it on the training data.

```
model = LogisticRegression()

model.fit(X_train, y_train)
```

Step 7: Make Predictions

Use the trained model to make predictions on the test data.

```
y_pred = model.predict(X_test)
```

Step 8: Evaluate the Model

Evaluate the model's performance using metrics like accuracy, confusion matrix, and classification report.

```
accuracy = accuracy_score(y_test, y_pred)

conf_matrix = confusion_matrix(y_test, y_pred)

class_report = classification_report(y_test, y_pred)


print(f'Accuracy: {accuracy}')

print('Confusion Matrix:')

print(conf_matrix)

print('Classification Report:')

print(class_report)
```

Step 9: Visualize the Results

Visualize the confusion matrix.

```
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```

CLASSIFICATION IN MACHINE LEARNING

Understanding Types, Algorithms, and Applications

Your Name

Date

INTRODUCTION TO CLASSIFICATION

Definition: Classification is a supervised learning technique used to predict the categorical class labels of new instances based on past observations.

Purpose: To categorize data into predefined classes or groups.

TYPES OF CLASSIFICATION

Binary Classification: Classifies data into two distinct classes.

Multiclass Classification: Classifies data into more than two classes.

Multilabel Classification: Each instance can belong to multiple classes simultaneously.

COMMON CLASSIFICATION ALGORITHMS

Logistic Regression: A linear model for binary classification problems.

Decision Trees: A tree-like model used for both classification and regression tasks.

Random Forest: An ensemble method that combines multiple decision trees.

Support Vector Machines (SVM): A powerful algorithm for binary classification tasks.

K-Nearest Neighbors (KNN): A simple, instance-based learning algorithm.

Naive Bayes: A probabilistic classifier based on Bayes' theorem.

Neural Networks: Deep learning models capable of handling complex classification tasks.

EVALUATION METRICS

Accuracy: The ratio of correctly predicted instances to the total instances.

Precision: The ratio of true positive predictions to the total positive predictions.

Recall: The ratio of true positive predictions to the total actual positives.

F1 Score: The harmonic mean of precision and recall.

Confusion Matrix: A table used to evaluate the performance of a classification model.

APPLICATIONS OF CLASSIFICATION

Spam Detection: Classifying emails as spam or not spam.

Sentiment Analysis: Determining the sentiment of text data (positive, negative, neutral).

Image Recognition: Identifying objects or people in images.

Medical Diagnosis: Predicting diseases based on patient data.

Fraud Detection: Identifying fraudulent transactions.

STEPS TO IMPLEMENT CLASSIFICATION IN PYTHON

1. IMPORT LIBRARIES
2. LOAD THE DATASET
3. EXPLORE THE DATA
4. PREPROCESS THE DATA
5. SPLIT THE DATA
6. TRAIN THE MODEL
7. MAKE PREDICTIONS
8. EVALUATE THE MODEL

LOGISTIC REGRESSION

```
import numpy as np

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load the dataset

data = pd.read_csv('your_dataset.csv')

# Preprocess the data

# Split the data

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train the model

model = LogisticRegression()

model.fit(X_train, y_train)

# Make predictions

y_pred = model.predict(X_test)
```

CONCLUSION

Summary: Recap the key points covered in the presentation.

Future Directions: Mention advanced topics like ensemble methods, deep learning for classification, and transfer learning.